

Assignment: Looping Statements in Python

Instructions

- Read each question carefully.
- Write the program by yourself.
- Use proper indentation.
- Do not copy code from others.
- Print output clearly.

Part 1: Basic Loop Practice

Q1. Print numbers from 1 to 20 using a loop.

- **Hint:** Use a `for` loop with `range()`.
- Control the starting and stopping values.

Q2. Print all even numbers from 1 to 50.

- **Hint:** Even numbers are divisible by 2.
- You may use modulus operator (%) or step value in `range()`.

Q3. Print numbers from 10 to 1 in reverse order.

- **Hint:** Use a negative step value in `range()`.

Part 2: Loop with Logic

Q4. Count how many numbers between 1 and 100 are divisible by 3.

- **Hint:** Use a counter variable.
- Increase counter when condition is true.

Q5. Ask user to enter a word and print each character one by one.

- **Hint:** Strings are iterable.
- A `for` loop can loop directly over a string.

Q6. Count how many vowels exist in a given string.

- **Hint:** Loop through each character.
- Use `if ch in "aeiouAEIOU"`.

Part 3: Multiplication and Pattern Thinking

Q7. Print multiplication table of any number entered by user.

- **Hint:** Loop from 1 to 10.
- Use formatted string printing.

Q8. Print the following pattern:

```
*
**
***
****
*****
```

- **Hint:** Use string multiplication.
- Loop from 1 to 5.

Part 4: While Loop Practice

Q9. Keep asking user for password until correct password is entered.

- **Hint:** Use `while True`.
- Use `break` when password is correct.

Q10. Create a countdown timer from 10 to 0.

- **Hint:** Decrease value inside loop.
- Stop loop when value reaches 0.

Part 5: Real-Life Simulation Questions

Q11. ATM PIN Verification System:

- User has only 3 attempts.
- Correct PIN is 1234.
- After 3 failed attempts, show "Card Blocked".
- **Hint:** Use a counter variable with a `while` loop.

Q12. Mobile Charging Simulation:

- Battery starts at 10%.
- Battery increases by 10% each loop.
- Stop when battery reaches 100%.
- **Hint:** Use condition-based `while` loop.

Challenge Question (Optional)

Q13. Guess the Number Game:

- Fixed number is 7.
- User keeps guessing until correct.
- **Hint:** Use `while True` and compare values.

Part 6: Additional Loop Practice

Q14. Print all numbers between 1 and 100 that are divisible by both 4 and 5.

- **Hint:** A number divisible by both 4 and 5 is divisible by their product.
- Use logical operator `and`.

Q15. Calculate the sum of all numbers from 1 to a number entered by the user.

- **Hint:** Create a variable to store sum.
- Add each number inside the loop.

Q16. Print all odd numbers between two numbers entered by the user.

- **Hint:** Use modulus operator to check odd numbers.
- Loop should start from the smaller number.

Q17. Reverse a number entered by the user.

- **Hint:** Use a `while` loop.
- Extract last digit using modulus (%).

Part 7: String-Based Loop Questions

Q18. Count how many spaces exist in a given sentence.

- **Hint:** Compare each character with a space.
- Increase counter when space is found.

Q19. Print only the consonants from a given string.

- **Hint:** Skip vowels using membership operator.
- Ignore spaces.

Q20. Check whether a given word is a palindrome.

- **Hint:** Reverse the string using a loop.
- Compare original and reversed strings.

Part 8: Real-Life Simulation (Advanced Thinking)

Q21. School Attendance System:

- Total students = 5.
- Ask attendance for each student (present/absent).
- Count total present students.
- **Hint:** Use loop and counter variable.